

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

```
In [2]: data = pd.read_csv("iris.data.csv", header=None)
data.columns = ['SepallLength', 'SepalWidth', 'PetalLength',
'PetalWidth', 'Species']
print("First 5 rows:")
print(data.head())
print("\nColumn Names:")
print(data.columns)
print("\nDataset Shape:")
print(data.shape)
```

First 5 rows:

	SepallLength	SepalWidth	PetalLength	PetalWidth	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Column Names:

```
Index(['SepallLength', 'SepalWidth', 'PetalLength', 'PetalWidth',
'Species'], dtype='object')
```

Dataset Shape:

```
(150, 5)
```

```
In [3]: X = data.iloc[:, :-1]
print("\nSelected Features:")
print(X.head())
```

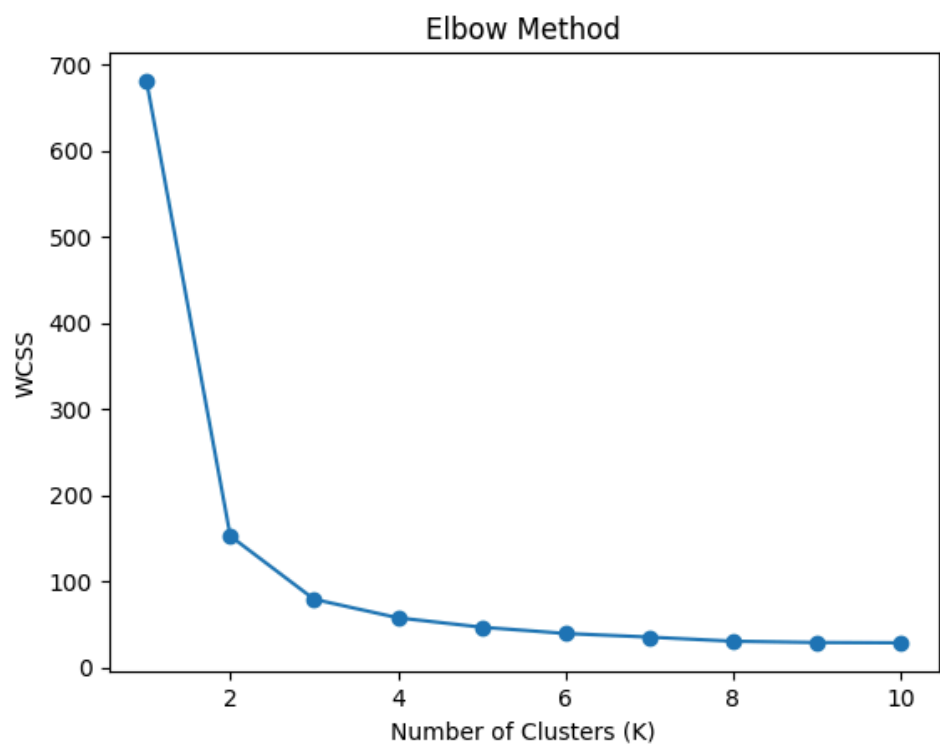
Selected Features:

	SepallLength	SepalWidth	PetalLength	PetalWidth
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
In [4]: wcss = []
for i in range(1, 11):
    kmeans_temp = KMeans(n_clusters=i, random_state=42)
    kmeans_temp.fit(X)
    wcss.append(kmeans_temp.inertia_)
print("\nWCSS Values:")
print(wcss)
```

```
WCSS Values:
[680.8243999999996, 152.36870647733915, 78.94506582597728,
57.44028021295475, 46.535582051282034, 39.251830892636775,
35.04275995246584, 30.217021122152712, 28.7564561965812,
28.424891802641817]
```

```
In [5]: plt.figure()
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.show()
```



```
In [6]: kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X)
print("\nK-Means Model Trained Successfully")
```

K-Means Model Trained Successfully

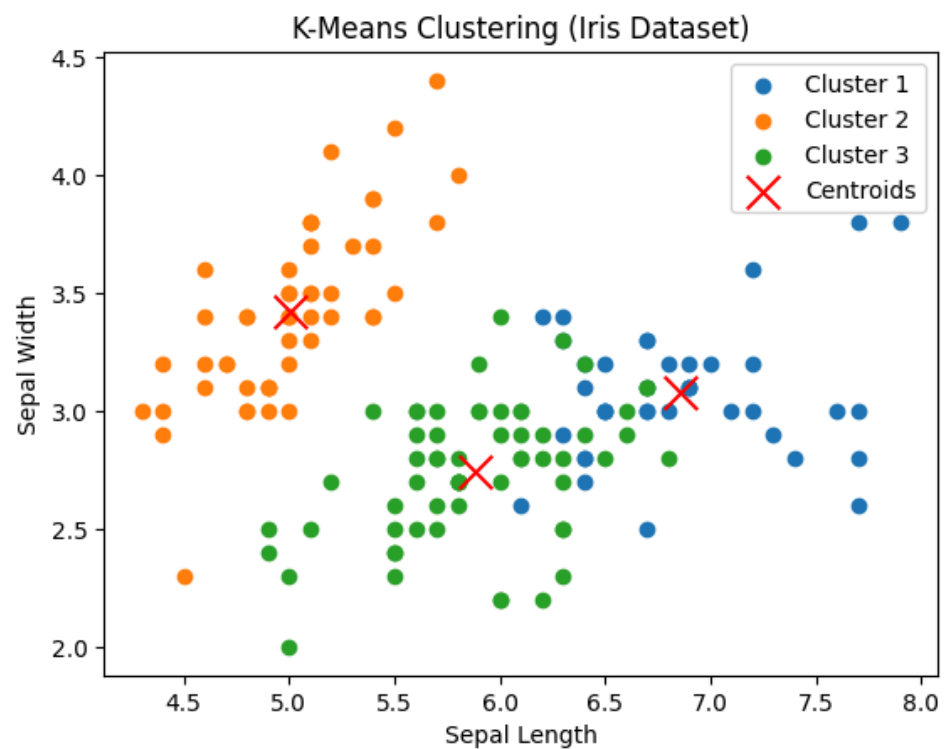
```
In [7]: kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X)
```

```
In [8]: data['Cluster'] = clusters
print("\nDataset with Cluster Labels:")
print(data.head())
```

Dataset with Cluster Labels:

	SepalLength	SepalWidth	PetalLength	PetalWidth	Species
Cluster					
0	5.1	3.5	1.4	0.2	Iris-setosa
1					
1	4.9	3.0	1.4	0.2	Iris-setosa
1					
2	4.7	3.2	1.3	0.2	Iris-setosa
1					
3	4.6	3.1	1.5	0.2	Iris-setosa
1					
4	5.0	3.6	1.4	0.2	Iris-setosa
1					

```
In [9]: plt.figure()
for i in range(3):
    plt.scatter(
        X.iloc[clusters == i, 0],
        X.iloc[clusters == i, 1],
        label=f'Cluster {i+1}'
    )
centers = kmeans.cluster_centers_
plt.scatter(
    centers[:, 0],
    centers[:, 1],
    c='red',
    marker='x',
    s=200,
    label='Centroids'
)
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("K-Means Clustering (Iris Dataset)")
plt.legend()
plt.show()
```



```
In [10]: print("\nCluster Centroids:")  
print(centers)
```

```
Cluster Centroids:  
[[6.85384615 3.07692308 5.71538462 2.05384615]  
 [5.006      3.418      1.464      0.244      ]  
 [5.88360656 2.74098361 4.38852459 1.43442623]]
```

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